



POM POWER

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POM HB SERIES

OUTDOOR HYBRID INVERTER
WALL MOUNTED TRANSFORMER FREE



Range: 3 KW to 5 KW (SINGLE PHASE)
8 KW to 50 KW (THREE PHASE)



Maximum Charging &
Discharging Current



Diesel Genset
Compatible



Best In Class
Switching Time



MPPT Current



Parallel
Upto 16 Units



Fanless Natural
Heat Dissipation

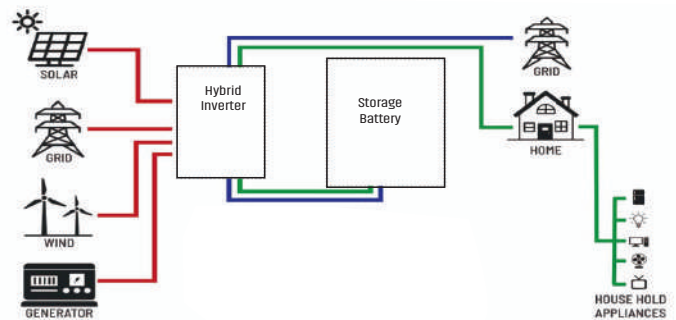
POM HB Series Hybrid inverters integrate solar and storage technologies with compact and durable Lithium (LFP) Batteries, combining the functions of a traditional solar inverter and a battery inverter in one device. The range of products are 3 kW to 50 kW and Paralleled up to 16 Times. This solution is ideal for homeowners seeking a solar power system with future upgrade potential, including battery storage.

Unlike a traditional solar grid-tie inverter, POM HB Series Hybrid inverter not only converts DC electricity from photovoltaic panels to AC for home use but also handles the bidirectional transfer of power to and from a LFP Battery storage system. One of the key advantages is the ability to draw power from the grid to charge the LFP Battery when solar generation is insufficient. The POM HB Series Hybrid inverter excels in DC-to-AC bi-directional power conversion, regulating power fluctuations and providing power monitoring capabilities. With Maximum Power Point Trackers (MPPT), it optimizes solar power output by adjusting voltage for efficient battery charging under varying conditions.

POM HB Series Hybrid inverter stands out for its seamless integration with the electricity grid. It allows excess solar energy to charge the LFP battery and transfer the surplus to the grid. This bi-directional power transfer capability distinguishes it from traditional solar grid-tie and battery inverters.

Crucially, during a power outage, POM HB Series Hybrid inverter paired with battery storage ensures continuous access to power, offering both off-grid and on-grid capabilities. Its design allows for easy retrofitting of battery storage, providing flexibility in adding components over time. Additionally, the inverter simplifies energy monitoring, consolidating data on performance and energy production in one accessible location.

POM HB Series Hybrid Inverters addresses issues like intermittent sunlight and unreliable grids, making it a robust solution. The advanced technology facilitates efficient power generation, management, and monitoring through net metering and Time-of-Day rates. Moreover, for those aiming to go off-grid, POM HB Series Hybrid inverter can manage solar panel operation, battery storage, and other power sources.



Features

- Lithium Battery Auto-restart Function, More Convenient For Lithium Battery Charging
- Utility Charging Voltage/PV Charging Voltage Adjustable, Match Different Battery Charging Requirements
- Battery Reverse Connection Protection with Fuse Switch, Safer Installation
- Support Working Without Battery: Reduce Solar System Cost
- High Precision of Output Voltage, $\pm 5\%$. Take Care of Your Appliances
- Intelligent Power Supply Mode. Intelligent Distribution of Solar Panel/Mains/Battery Power Shares
- Slim Body. Convenient Installation and Transportation
- PF 1.0, High Efficiency, Lower Consumption, Energy Conservation/ Environmental Protection/ Electricity Saving/ Cost Saving
- Parallel Function Up to Maximum 16 Units: Enlarge More Loads
- Communication Option: External WIFI, Supervise at Any Time

INVERTERS

FROM POM POWER



TECHNICAL SPECIFICATIONS

Product Model	POM 301 HB	POM 501 HB	POM 801 HB	POM 1001 HB	POM 1201 HB
INPUT DC(PV)					
Recommended max.PV power	5kW	8kW	12 kW	15kW	18kW
Max.input voltage	580V			1000V	
Rated input voltage	350V			600V	
Start-up voltage	120V			200V	
MPPT voltage range	90-520V	120-520V		200-850V	
Max.input current	20A/20A			26A/26A	
Max.short circuit current	22.5A/22.5A			39A/39A	
MPPT number/Max.input strings number	2/2			2/4	
INPUT/OUTPUT DC(BATTERY)					
Battery type	Lead-acid/Li-ion				
Battery voltage range	42-58V				
Max.charge/discharge power	3kW	5kW	8kW	10kW	12kW
Max.charge/discharge current	72A	120A	30A		
Communication Mode	RS485/CAN				
OUTPUT AC					
Rated output power	3kW	5kW	8kW	10kW	12kW
Max.output power	3.3kW	5.5kW	8.8kW	11kW	13.2kW
Max.apparent output power	3.3kVA	5.5kVA	8.8kVA	11kVA	13.2kVA
Rated output current	13.6A	22.7A	12.1A	15.2A	18.2A
Max.output current	15.1A	25.2A	13.4A	16.7A	20A
Switch time of grid-connected and off-grid	<10ms				
Rated grid output voltage/frequency	L/N/PE 220V/230V 50Hz/60Hz		3L/N/PE 380V/400V 50Hz/60Hz		
Power factor	0.8 leading ...0.8 lagging				
THDv(@linear load)	<3%				
THDi	<2%				
INPUT AC					
Max.apparent input power	3.3KVA	5.5KVA	8.8kVA	11 kVA	13.2kVA
Max.input current	15.1A	25.2A	13.4A	16.7A	20A
Rated grid input voltage/frequency	L/N/PE 220V/230V 50Hz/60Hz		3L/N/PE 380V/400V 50Hz/60Hz		
EFFICIENCY					
Max.efficiency	97.50%			98.10%	
STANDARD					
Safety	IEC62109-1/-2				
EMC	EN61000-6-1/EN61000-6-3		EN61000-6-1/EN61000-6-3		
BASIC PARAMETERS					
Dimensions (W*H*D)	362mmx546mmx252mm			556mmx482mmx196mm	
Cooling concept	Natural				
Weight	21.5kg			27.5kg	
Degree ofprotection	IP66				
Operating ambient temperature	-25℃~60℃ (>45 derating)				
Max.operation attitude	3000m				
Communication Mode	RS485/WIFI/GPRS/LAN				

TECHNICAL SPECIFICATIONS

Product Model	POM 801 HB	POM 1001 HB	POM 1201 HB	POM 1501 HB	POM 2001 HB
Input DC(PV)					
Recommended max.PV power	12 kW	15kW	18kW	22.5kW	30kW
Max.input voltage	1000V				
Rated input voltage	620V				
MPPT voltage range	200-850V				
Continuous input current	32A/32A				
MPPT number/Max.input strings number	2/4				
Batteries					
Input/Output DC(Battery)	Lead-acid/Li-ion				
Battery voltage range	120-600V				
Max.charge/discharge power	8kW	10kW	12kW	15kW	20kW
Max.charge/discharge current	50A				
Communication Mode	RS485/CAN				
Charging Curve	Three-paragraph				
AC (Off grid)					
Max. output power	8kW	10kW	12kW	15kW	20kW
Switch time of grid-connected and off-grid	<10ms				
THDi	<2% (Linear load)				
AC (On grid)					
Rated output power	8kW	10kW	12kW	15kW	20kW
Max. output current	13.4A	16.7A	20A	25A	33.3A
Max. input current	13.4A	16.7A	27.3A	34.1A	45.5A
Rated grid output voltage	3L/N/PE, 380V/400V				
Rated grid output frequency	50Hz/60Hz				
Power factor	>0.99(0.8 leading ...0.8 lagging)				
THDi	<2% (Full load)				
Efficiency					
Max.efficiency	> 97.5%				
MPPT efficiency	> 99.5%				
PROTECTION FUNCTION					
Island protection	Available				
Input Protection for reverse connection	Available				
100% three-phase unbalanced load	Available				
AFCI protection	Optional				
SYSTEM DATA					
Maximum number of parallel machines	16				
Communication Port	RS485/WIFI(Optional)/4G(Optional)/LAN(Optional)				
Dimensions	567mmx497mmx237mm		554mmx526mmx206mm		
Weight	27.5kg		29kg		31.6kg
Operating ambient temperature	-40 ~ +60°C (>45 derating)				
Degree of protection	IP66				
Cooling Concept	Natural			Intelligent air-cooled	

TECHNICAL SPECIFICATIONS

Product Model	POM 2501 HB	POM 3001 HB	POM 3301 HB	POM 5001 HB
Input DC(PV)				
Recommended max.PV power	25 kW	30 kW	33 kW	50kW
Max.input voltage	1000V			
Rated input voltage	620V			
MPPT voltage range	200-850V			
Continuous input current	32A/32A		36A/36A	
MPPT number/Max.input strings number	2/4			
Batteries				
Input/Output DC(Battery)	Lead-acid/Li-ion			
Battery voltage range	200-800V		160-800V	
Max.charge/discharge power	25 kW	50 kW	33 kW	50kW
Max.charge/discharge current	50A			
Communication Mode	RS485/CAN			
Charging Curve	Three-paragraph			
AC (Off grid)				
Max. output power	25 kW	30 kW	33 kW	50kW
Switch time of grid-connected and off-grid	<10ms			
THDi	< 3%			
AC (On grid)				
Rated output power	25 kW	30 kW	33 kW	50kW
Max. output current	41.7	50	55	72.5 A
Max. input current	41.7	50	55	79.7A
Rated grid output voltage	3L/N/PE, 230 (400)			
Rated grid output frequency	50Hz/60Hz (45-55 Hz/55-65Hz)			
Power factor	0.8 leading ...0.8 lagging			
THDi	<3 %			
Efficiency				
Max.efficiency	>98.2 %			
MPPT efficiency	>99.9 %			
Protection Function				
Island protection	Integrated			
Input Protection for reverse connection	Integrated			
100% three-phase unbalanced load	Integrated			
AFCI protection	Integrated			
System Data				
Maximum number of parallel machines	16		12	
Communication Port	WIFI//LAN (Optional)			
Dimensions	640mmx490mmx290mm			940mmx595mmx-340mm
Weight	48			55
Operating ambient temperature	-30 ~ +60 °C			
Degree of protection	IP66			IP65
Cooling Concept	Smart Fan Cooling			



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Grid Renewable Energy Storage Systems (GRES) Battery Energy Storage

Systems (BESS)
Standalone UPS
Modular UPS
Rackmount UPS

Micro Data Center

Modular Data Center

EV-DC Fast Chargers

Solar On-grid Inverters

Hybrid Inverters

Off-Grid Inverters

Solar Power Plants

Servo Controlled
Voltage Stabilizers

Frequency Converters

Isolation Transformers



SYSTEMS & SERVICES PVT.LTD.

FACTORY & HEAD OFFICE

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15, Kasturba Gandhi Marg
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TRIVANDRUM

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Kudapanakunnu P. O
Thiruvananthapuram - 695 043
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PUNE

Plot no P 81, D 2 Block
Behind Kabre Hospital
MDC Chinchawad, Pune - 411 038
Mob. 9036180744

HUBBALLI

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Mob. 9036180742

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TIRUPATI

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HYDERABAD

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MUMBAI

No. 620, Marathon Max, Mulund Goregaon
Link Road, Mulund West, Mumbai - 400 080
Mob. 9035032568

VIJAYAWADA

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Mob. 9902228907

PATNA

Near Shyama Villa Apartment,
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Patna-800001
Mob. 9606978032

LUCKNOW

451/6 Indira Nagar, A-Block
Lucknow-226016
Mob. 9606978038

CHANDIGARH

05, Rajpur Khurd
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Chandigarh-160003
Mob: 9902226370

CHENNAI

#2, Mukhtlal Naidu Street
Rear portion of first floor, Little mount
Chennai-600015
Mob. 9108992485

COIMBATORE

Skyon, # 451, 1st Floor, Kamarajar Road,
Peelamedu, Coimbatore - 641004
Mob: 9036180749

KOLKATA

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ERNAKULAM

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